CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. Examination (IT) & Second Semester of M. Tech. Examination (CE)
Dec 2013-Jan 2014

CE712 Software Project Management & Quality Assurance (SPM & QA)

Date: 07.01.2014, Tuesday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Rough work is to be done in the last page of main supplementary, please don't write anything on the question paper.
5. Indicate clearly, the option(s) you attempt along with its respective question no.
6. Figures to the right indicate marks.

SECTION-I

Q-1 **Answer the following questions.**

1. How might **Project Characteristics** affect your choice of the **Process Model** and **Team Selection**? How might customer involvement affect the choice of the **Process Model**? 4
2. Discuss the techniques for estimating ***project duration*** and ***determining the staffing pattern***. 4
3. Explain the significance of ***calendar control*** in **MS Project Software**. 3

Q-2

[A] Which are the characteristics used by **Project Manager** for doing *resource allocation, monitoring* and *controlling* the progress of the system? 4

[B] Software ***project planning*** entails which activities? What are the difficulties faced in measuring the ***software costs***? 4

OR

[B] What is **Software Quality Audit**? What is the goal of it? Point out. 4

[C] What is a **risk**? If you are the **Project Manager** of a large software development project, what steps would you follow to manage **risks** in your **software projects**? What are **risk management activities**? 4

Q-3

[A] What are the things to be considered before using a ***software package*** for **Project Management**? 4

[B] What can be done, if *scope* and *details* of **Software Projects** are changed frequently? How can a **Project Manger** minimize their impacts of it on **Projects**? 4

OR

[B] Discuss the importance of a good ***time management***. How **activity sequencing** is done? 4

[C] What do you mean by **Work Break –down Structure** (WBS) in the context of a software projects? Also discuss in brief about **Milestone Chart**. 4

SECTION-II

Q-4

1. What are the needs of **Project Management Methodologies**? Explain the benefits of it. **3**
2. What do you mean by ***software quality*** and ***quality software***? Discuss various software **quality assurance activities**. **4**
3. When during the software life cycle are ***verification*** and ***validation*** performed? Can one be used in place of the other? **4**

Q-5

[A] Describe the **software effort** and **cost estimation** techniques in detail. **4**

[B] Which are the **methodologies (approaches)** to design the ***test cases***? Explain in brief. What is a **test suite**? How to design ***optimal test suite***? **4**

[C] Discuss the ***graphical notations*** which are used to illustrate the **Project Schedule**. **4**

OR

Q-5

[A] Explain the difference between **Critical Path Analysis** and **Critical Chain Schedule**? Is **critical path** important if *only one person* is working on a software project? Discuss the concept of **PERT/CPM** in defining *optimum schedule*. **4**

[B] Justify that **FPs** give better *estimate* of software *project size* than **LOC**. **4**

[C] There are a number of unique **challenges** and **elements** that are involved in *leading* and *managing* **software engineering** and **IT projects.** What are the unique **challenges** and **elements** that are involved in managing IT **Projects**? Learning effective **Project Management** techniques will allow you to complete your *projects* on **time** and bring them in under **budget.** Do you **agree** with this statement? **Justify** your answer. **4**

Q-6

[A] Explain in detail, different ***levels*** of **testing**. Also mention *who* is responsible for carrying out that **level of testing** and *where*? **4**

OR

[A] In the present scenario of **frequent failure(s)** of software projects, can you put forward some **strategies** that facilitate successful **software projects**? What are the various **measures** for failure of **software projects**? **4**

[B] Explain the necessity of **SCM** (Software Configuration Management). Why is **SCM** crucial to the success of a **large software Project**? Explain about **SCM** *lifecycle* and *requirement change management process.* **4**

[C] How to start and keep the **projects** on **track**? How to **monitor** and **control project progress**? How to **learn** effective **project management techniques**? What are the *advantages* of the same? **4**

OR

[C] What are the five levels of **CMM**? Is it possible for an organization to achieve a *higher level* of **CMM** without achieving a *lower one*? **Justify** the statement. **4**